A Bathymetry Mapping Approach Combining Log-Ratio and Semianalytical Models Using Four-Band Multispectral Imagery Without Ground Data

Haoyang Xia, Xiaorun Li, Huaguo Zhang, Juan Wang, Xiulin Lou, Kaiguo Fan, Aiqin Shi, and Dongling Li

***Abstract*— Four-band multispectral remote sensing imagery is widely used for a variety of purposes and has a long historical data record. However, most of the existing bathymetry inversion methods are either unable to determine the optimal solution through theoretical or semianalytical models due to the limited number of bands, or their application is limited by the difficulty in obtaining *in situ* data. In this article, a log-ratio model and a semianalytical model are combined to develop a new shallow water depth inversion method (L-S model) using four-band multispectral remote sensing images without the need for supporting truth data. A case study was conducted for Ganquan Island in the South China Sea, using four-band multispectral imagery from the GeoEye-1, WorldView-2 (four bands selected), Sentinel-2 (four bands selected), and Gaofen-1 satellites; a sound range of 30 m was achieved. When compared to LiDAR-measured water depth data, GeoEye-1, WorldView-2, Sentinel-2, and Gaofen-1 data have root-mean-square errors (RMSEs) of 1.33–1.97 m. In addition, compared with the results of the log-ratio model trained using 200 LiDAR-based depth readings, the L-S model results obtained for the four satellite types are similar in the RMSE. These results show that the L-S model can achieve results that are close to those of the log-ratio model without the need for external inputs. This provides a feasible new method for bathymetry inversion in areas without truth data using only four-band imagery.**

***Index Terms*— Bathymetry mapping, four-band imagery, log-ratio model, no ground data, semianalytical model.**

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I. Introduction

BATHYMETRY of shallow water areas provides valuable information about habitats and topography for a variety of end uses, including biological monitoring and navigation safety [1]. Compared to traditional airborne or ship-borne surveys, satellite observations provide higher spatial coverage and temporal resolution at lower cost [2]. In particular, multispectral remote sensing images can be obtained from a large number of satellite sources with a long historical archive, including Landsat-5, Landsat-7, Landsat-8, IKONOS, QuickBird, GeoEye-1, WorldView-2/3/4, Gaofen-1/2, and Sentinel, among others. Visible wavelengths are useful for bathymetry mapping; however, many high-resolution satellites have only three visible bands: usually red, green, and blue. In recent decades, many successful bathymetry mapping algorithms that use multispectral or hyperspectral imagery have been developed. Working on the premise that the water environment is approximately constant, statistical models are created by determining the statistical relationship between the radiance values obtained from a remote sensing image and *in situ* bathymetry data. In the simplest single-band models, the radiance received by the optical remote sensor is expressed as the sum of the radiance from the deep-water region and the reflected radiance from the seabed. When this model is simplified, an analytical relationship between the radiance and the depth can be expressed using a regression of only two empirical parameters [3], [4]. Assuming that the reflectance ratios of two bands on different substrate types are constant, a log-linear model is established, which can also be extended to multiple bands [5]. Furthermore, the log-ratio model using two bands cannot produce a negative difference between the radiance received by an optical remote sensor and the radiance in deep water, making it more stable and accurate [6]. Inversion accuracy can be further improved by the introduction of factors such as suspended sediment measurements into the statistical models [7]. At the top of the complexity hierarchy are artificial neural network water depth inversion algorithms, which use self-learning, self-organization, self-adaptation, and nonlinear dynamic processing, and provide more accurate results than the traditional statistical methods [8]–[10]. However, these models rely on a little amount of *in situ* data [6], [11], which limits their use and, in particular, artificial neural

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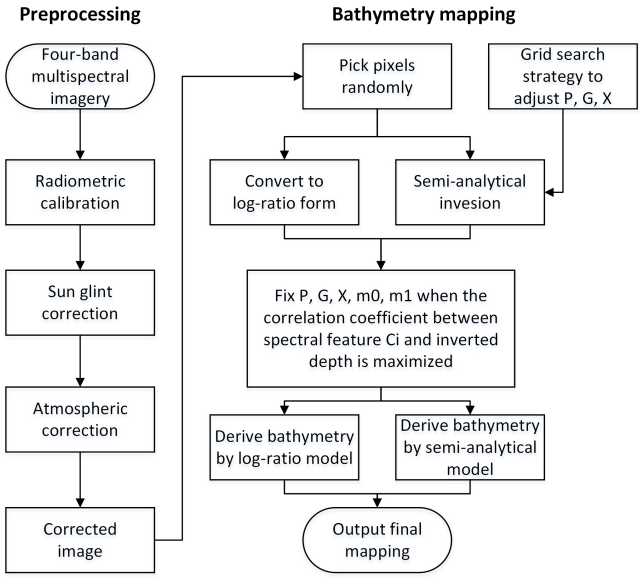


Fig. 1. Flow diagram of the L-S model.

networks require a large amount of data. The hyperspectral optimization process exemplar (HOPE) model is the most influential and widely used hyperspectral remote sensing water depth inversion model [12], [13]. This model calculates the relationship between the remote sensing reflectance of shallow water, water column inherent optical properties (IOPs), bathymetry, and bottom albedo [14]. It fully considers the absorption and scattering factors of several water components, and the physical mechanism is relatively complete. The major advantage of the HOPE model over others in this category is that it can directly conduct depth inversion without *in situ* water depth training data [15]–[17]. The HOPE model was originally designed for use with hyperspectral remote sensing imagery, though updated versions can also use eight-band multispectral data [18]. The HOPE model requires enough equations, and therefore four-band multispectral remote sensing images cannot be used directly in this model.

Recently, Chen *et al.* [19] proposed a dual-band algorithm that achieved good results without using any ground data. This algorithm first formulates a log-linear model and then solves for all unknown model parameters by sampling different types of pixels from those directly extracted from multispectral images. However, the process of sampling pixels brings uncertainties in practice.

In this article, we propose a comprehensive approach combining a log-ratio model and a semianalytical model, which we call the L-S model, and which can be used successfully with most four-band multispectral remote sensing imagery methods. We tested this method using four types of remote sensing imagery methods with different spatial resolutions to calculate bathymetry in the area of Ganquan Island in the South China Sea and verified the results using LiDAR data.

II. Methods

The workflow of the proposed L-S model, including some preprocessing, is shown in Fig. 1. One limiting factor when using visible imagery for bathymetry inversion is sun glint—the specular reflection of the sun on the rough water surface—which we remove algorithmically [20]. Another factor that cannot be ignored is absorption and scattering by particles in the atmosphere, which is more significant in ocean remote sensing, because the signal from the sea surface is usually very small [21]. After applying these corrections, we apply the L-S method, described in detail below. Because the semianalytical model requires at least six equations, and therefore six bands, we first make the simplifying assumption that water IOPs are nearly uniform within a small geographic region compared to changes in other relevant variables, and then use a standard log-ratio model to determine the parameters of water properties. In summary, a complete four-band multispectral imagery bathymetry mapping method has been developed, which is documented as Fig. 1.

A. Sun Glint Correction

Sun glint on rough water surfaces is a strong disturbance source and must be effectively suppressed. Therefore, methods for eliminating the impact of sun glint are an important research topic. In this article, we make use of the linear relationship between the reflectance of near-infrared (NIR) and visible bands [20] to eliminate the effects of sun glint.

The corresponding mathematical relationship is as follows:

$$\rho_{\mathrm{TOA}}^{\mathrm{deg}}(\lambda) = \rho_{\mathrm{TOA}}(\lambda) - b_{\lambda}(\rho_{\mathrm{NIR}} - \mathrm{MIN}\rho_{\mathrm{NIR}}) \tag{1}$$

where $\rho_{\mathrm{TOA}}^{\mathrm{deg}}(\lambda)$ is the surface reflectance after the sun glint is removed for a given visible band, $\rho_{\mathrm{TOA}}(\lambda)$ is the reflectance at the top of atmosphere (TOA), ρ_{NIR} is the NIR reflectance, $\mathrm{MIN}\rho_{\mathrm{NIR}}$ is the minimum NIR reflectance for a given scene, and b_{λ} is the regression constant relating the visible and NIR reflectances in the scene. λ is the center wavelength of different bands.

B. Atmospheric Correction

In ocean remote sensing, a large portion of energy received by satellite sensors comes from the atmosphere. Therefore, in order to achieve more accurate water depth inversion results, an atmospheric correction must be performed. There are many mature atmospheric correction algorithms, such as the fast line-of-sight atmospheric analysis of hypercubes (FLASSH) module, the dark pixel method based solely on information within the image, and the Second Simulation of the Satellite Signal in the Solar Spectrum (6S) model that simulates the solar radiation transmission process in the atmospheric system and calculates the radiance retrieved by the satellite [22].

An accurate atmospheric correction is vital for accurate bathymetry inversions; however, determining the composition of the scattering and absorbing components of the atmosphere can be difficult. The easiest way to obtain atmospheric aerosol parameters is to extract them directly from the image. In this article, we chose the dark pixel method for atmospheric correction, which does not require explicit knowledge of atmospheric aerosol content, and compared the results with the MODIS reflectance product.

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C. Semianalytical Model

The core of the water depth estimation method based on the semianalytical model and nonlinear optimization is to establish a functional relationship between the water surface remote sensing reflectance—R_{rs}, seafloor albedo, water depth, and IOPs. According to [11], [13], and [18], for multispectral imagery, the modeled water surface remote sensing reflectance R_{rs}^{m} can be formulated as follows:

$$R_{\mathrm{rs}}^{m}(\lambda) = f(P, G, X, B, H, Y, \lambda) \tag{2}$$

where f is a multivariable nonlinear equation, P is the phytoplankton absorption coefficient at 440 nm, G is the sum of Gelbstoff and Detritus absorption at 440 nm, X is the particle backscattering coefficient, Y is the spectral shape parameter, B is the bottom albedo, H is the water depth, and λ is the central wavelength of different bands.

The specific radiative transfer equations are as follows:

$$R_{\mathrm{rs}}^{m} = \frac{0.5 r_{\mathrm{rs}}}{1 - 1.5 r_{\mathrm{rs}}} \tag{3}$$

$$r_{\mathrm{rs}} = r_{\mathrm{rs}}^{C} + r_{\mathrm{rs}}^{B} \approx r_{\mathrm{rs}}^{\mathrm{dp}} \left(1 - e^{-\left[\frac{1}{\cos(\theta_{\omega})} + D_u^{C}\right] kH}\right) + \frac{\rho}{\pi} e^{-\left[\frac{1}{\cos(\theta_s)} + \frac{D_u^{B}}{\cos(\theta_v)}\right] kH} \tag{4}$$

$$r_{\mathrm{rs}}^{\mathrm{dp}}(\lambda) \approx (0.084 + 0.170u(\lambda))u(\lambda) \tag{5}$$

$$u(\lambda) = \frac{b_b(\lambda)}{a(\lambda) + b_b(\lambda)} \tag{6}$$

$$k(\lambda) = a(\lambda) b_b(\lambda) \tag{7}$$

$$D_u^{C}(\lambda) \approx 1.03(1 + 2.4u(\lambda))^{0.5} \tag{8}$$

$$D_u^{B}(\lambda) \approx 1.10(1 + 5.0u(\lambda))^{0.5} \tag{9}$$

$$b_b(\lambda) = b_{bw}(\lambda) + \varepsilon(\lambda) b_{bp}(\lambda) \tag{10}$$

$$a(\lambda) = a_w(\lambda) + a_{\phi}(\lambda) + a_g(\lambda) \tag{11}$$

$$\varepsilon(\lambda) \approx 1 + \left[0.1 + \frac{0.8 b_{bp}(\lambda)}{b_b(\lambda)}\right] \sin(\theta_s)\sin(\theta_v) \tag{12}$$

$$\alpha_{\phi} = [\alpha_0 + \alpha_1 \mathrm{In}(P)]P \tag{13}$$

$$\alpha_g(\lambda) = G \exp[-0.014(\lambda - 440)] \tag{14}$$

$$b_{bp}(\lambda) = X\left(\frac{400}{\lambda}\right)^{Y} \tag{15}$$

$$\rho(\lambda) = B\rho_{\mathrm{norm}}(\lambda) \tag{16}$$

$$\mathrm{MIN}_{\mathrm{optimization}} = \sum_{ch_1}^{ch_n} \left\{R_{\mathrm{rs}}^{m}(ch) - R_{\mathrm{rs}}(ch)\right\}^2 \tag{17}$$

where r_{rs} is the modeled subsurface reflectivity, $r_{\mathrm{rs}}^{\mathrm{dp}}$ is the reflectivity for optically deep water, D_u^{C} and D_u^{B} are the optical path-elongation factors for scattered photons from the water column and the bottom, respectively, k is the diffuse attenuation coefficient, H is the water depth, θ_s is the subsurface solar zenith angle, and θ_v is the subsurface view angle.

By nonlinear optimization (17), these six parameters P, G, X, B, H, and Y can be derived simultaneously. Note that with the decrease of the number of bands, the accuracy of the semianalytical model also decreases.

D. Log-Ratio Model

The log-ratio method relies on the fact that absorption by water varies with the wavelength. Generally speaking, as the water depth increases, the water leaving reflectance also decreases. However, the attenuation degree of the short-wavelength band is less than that of the long-wavelength band, which enables the log ratio of reflectance of different bands have an effective correlation with the water depth. The results in [17] show that the changes in albedo affect different bands similarly, and the changes in the log ratio of the bands are even smaller. The corresponding mathematical relationship is as follows:

$$z = m_1 \frac{\ln(n R_w(\lambda_i))}{\ln(n R_w(\lambda_j))} - m_0 \tag{18}$$

where m_1 and m_0 are tunable constants controlled by *in situ* data and n is a fixed constant for all areas. In this article, the value of n is fixed to 10 000 to ensure a positive logarithm under any condition. λ_i is the central wavelength of the blue band, while λ_j is the central wavelength of the green band.

E. L-S Model

In the semianalytical model, the parameters P, G, X, B, and Y, and the water depth, H, determine the remote sensing reflectance spectrum. By using a nonlinear optimization technique, such as the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm [23], we can invert the bathymetry when the relative error is minimized.

Since the number of equations (bands) is too small when using four-band multispectral imagery, the above-mentioned method cannot obtain the nonlinear optimization solution directly. In order to enable the semianalytical model to be applied to the four-band remote sensing imagery, we must make some simplifying assumptions. The six parameters can be divided into three categories, namely geographical environmental factors (B, H), water column IOPs (PG, X), and spectral character Y. Relative to the geographical environmental factors (B, H) and the spectral parameter (Y), the change in water properties should not vary significantly within a small geographic area. Therefore, P, G, and X can be regarded as a near-constant in clear water.

The next step is to determine the values of P, G, and X. Compared to the direct use of hyperspectral imagery for water depth inversion, the information content of a four-band image is greatly reduced. To compensate for this issue, we use the log-ratio model to link different pixels in the same image, because as per our assumptions, water IOPs are nearly uniform within the given region [24]. In the process of water depth inversion in traditional semianalytical models, different pixels are independent of each other. That is, they are processed in parallel during the inversion process. However, the mutual information between them is ignored, and we use the following methods to ensure the success of the inversion.

1) Pick hundreds of pixels randomly as self-infer points (SIPs).
2) Set the initial values of P, G, and X (for instance: $P = 0.005$, $G = 0.005$, and $X = 0.004$).

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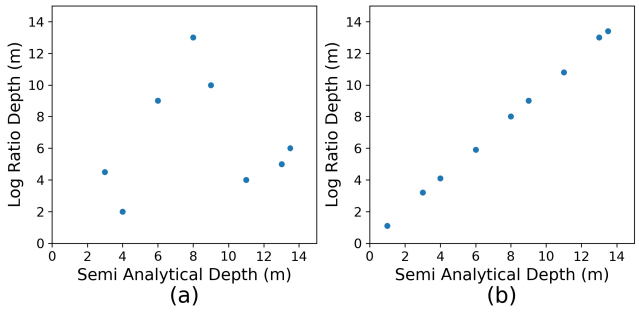


Fig. 2. (a) Typical scatter plot of semianalytical depth and log-ratio depth when P, G, and X are randomly guessed. (b) Scatter plot when the correlation coefficient between semianalytical depth and C_i (19) is maximized; at this time, P, G, and X are considered correct.

3) Use the semianalytical model to invert the other three parameters B, H, and Y for those initial values of P, G, and X.
4) We now have semianalytic inversion results for each point, which are equal to the inversion results obtained from the log-ratio method, and can be expressed as

$$\begin{aligned} \text{Semianalyticaldepth} &= \text{Log-ratio depth} \\ &= m_1 * C_i + m_0 \qquad (19) \\ \text{where } C_i &= \frac{\ln(nR_w(\lambda_i))}{\ln(nR_w(\lambda_j))}. \qquad (20) \end{aligned}$$

For each SIP, C_i is different, but certain—it is determined by the reflectance at each point. We now have hundreds of equations like (19) with only two unknowns. Using the least-squares method, the values of m_1 and m_0, as well as the log-ratio derived depth of each point, can be fit. We can then compare the semianalytical water depth and the log-ratio water depth belonging to the same point using a scatter plot. A typical scatter plot, with random values of P, G, and X, is shown in Fig. 2(a).
5) Adjust the values of P, G, and X until the correlation coefficient between the semianalytical depth and C_i (20) is maximized. At this point, the P, G, and X values can be considered to correctly reflect the water column information. The scatter plot at this time is shown in Fig. 2(b).
6) As P, G, and X of the semianalytical model and m_0, m_1 of the log-ratio model are now known, we can invert the water depth using either the semianalytical model or the log-ratio model.

It should be noted that this is a brute search algorithm. This method has two advantages: 1) it can find the global optimal solution and 2) it facilitates parallel calculation. The ranges for P, G, and X were very important in this article; these were determined using the database of the IOCCG report [25]. The steps of P, G, and X are empirical values, which are determined by balancing the precision requirement and the calculation pressure. All the parameter range settings were shown in Table I. The key pseudocodes are shown in Algorithm I.

TABLE I
PARAMETERIZATION AND OPTIMIZATION RANGE FOR L-S MODEL

Parameters	Initial value	Range	Step
P	0.005	0.00075-0.1	0.005
G	0.005	0.005567-0.03	0.005
X	0.004	0.003926-0.15	0.01
B	0.5	0.002 - 1	
H	3	0 - 30	
Y	1	0 – 2.5	

Algorithm 1 L-S Algorithm

Require: the initial value of P, G, X: *Pinit*, *Ginit*, *Xinit*, the range of P, G, X: *Prange*, *Grange*, *Xrange*, the step of P, G, X: *Pstep*, *Gstep*, *Xstep*, the set of chosen points *SIPset*
Ensure: the value of P, G, X should be in their respective ranges
1: *Pset* = generateSet(*Pinit, Prange, Pstep*);
2: *Gset* = generateSet(*Ginit, Grange, Gstep*);
3: *Xset* = generateSet(*Xinit, Xrange, Xstep*);
4: *Cset* = calcLogRatio(*SIPset*)
5: correlation = 0
6: **for** *P* in *Pset* **do**
7: **for** *G* in *Gset* **do**
8: **for** *X* in *Xset* **do**
9: *Bset, Hset, Yset* = HOPE(*P, G, X, SIPset*)
// through BFGS optimization algorithm
10: *temp* = calculateCorrelationCoefficient(*Cset, Hset*)
11: **if** *temp* > *correlation* **then**
12: *correlation* = *temp*
13: *Hbest* = *Hset*
14: *Pbest* = *P*
15: *Gbest* = *G*
16: *Xbest* = *X*
17: **end if**
18: **end for**
19: **end for**
20:**end for**
21:m_1, m_0 = calculateCoefficient(*Cset, Hset*)
//through linear regression method
22:**Return** m_1, m_0

III. TESTS AND RESULTS

A. Test Site

The L-S model was tested on water surrounding the Ganquan Island in the South China Sea, which is covered by sand, submerged vegetation, and coral reefs. The center coordinates of the site are 16°48′N and 111°59′E. In terms of morphodynamics, the site is a static region without human activity. In this initial test case, we used GeoEye-1 data as an example to illustrate the L-S model. Later, we look at the results using other satellites to test the robustness of the algorithm. As shown in Fig. 3, four points (A, B, C, and D) are used to examine the result of atmospheric correction, with the spectrum of deep water acquired by MODIS as a reference.

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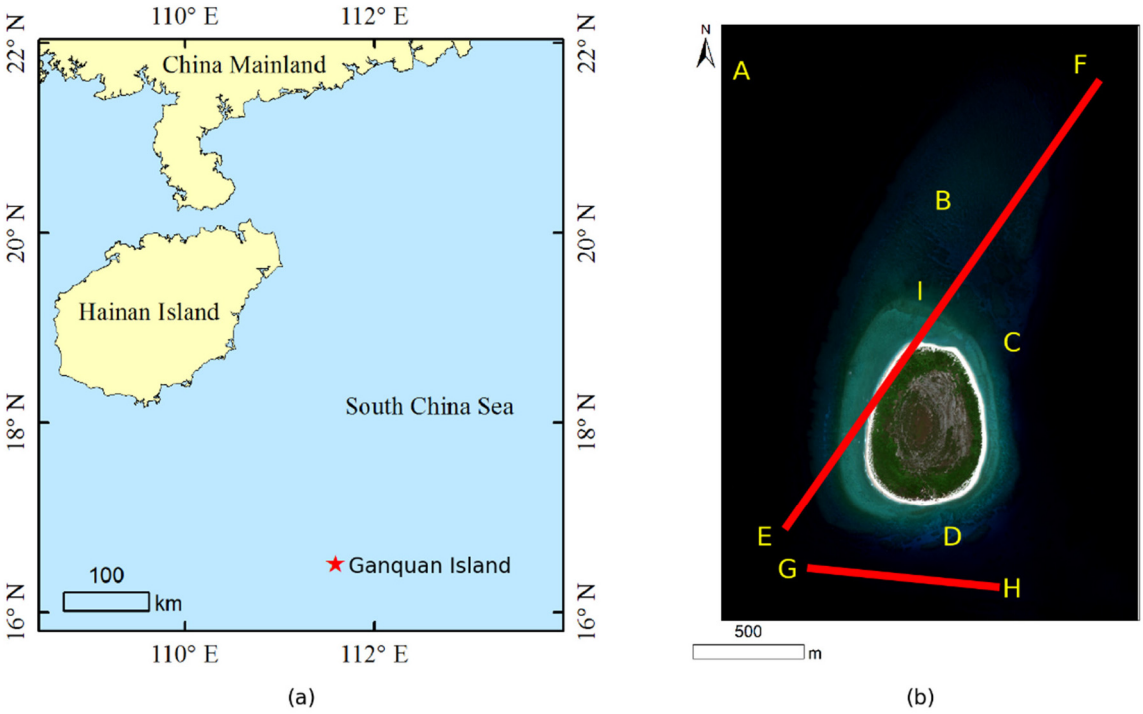


Fig. 3. (a) Location of Ganquan Island. (b) True color image of the island with test data points and transects.

The transect GH is used to verify the effect of the sun glint correction and the EF is used to determine which pixels are appropriate for the use of the L-S model for bathymetry inversion.

B. LiDAR Data

In this article, we used LiDAR data to validate the results of the L-S model. The LiDAR data were obtained using the Mapper5000 system on September 23, 2017; the original sampling interval is 1.1 m; and the depth accuracy is 0.23 m [27]. In order to facilitate the comparison with remote sensing image results having different resolutions, the resolution of LiDAR data used was interpolated to 10 m using only data points that were entirely under water.

C. Test Case Using GeoEye-1 Imagery

The GeoEye-1 satellite was launched on September 6, 2008 and the four bands (blue, green, red, and NIR) used in this article have a resolution of 2.0 m.

The GeoEye-1 image used in this test case was acquired on July 2, 2015. For this image, the solar zenith angle was 22.7°, the azimuth angle was 69.2°, the satellite zenith angle was 24.9°, and the azimuth angle was 92.6°.

As described in Section II, we first performed the sun glint and atmospheric corrections. As shown in Fig. 4(d), after the sun glint correction, the reflectivity image is visually clearer and the changes in reflectivity are smoother. In this scene, after application of the sun glint correction, the average blue band reflectance in deep water decreases from 0.1117 to 0.1105 (−1.1%), the green band reflectance decreases from 0.0687 to 0.0674 (−1.9%), and the red band reflectance decreases from 0.0347 to 0.0333 (−4.1%).

After the sun glint correction, the dark pixel atmospheric correction method was applied. As shown in Fig. 4(e), the spectrum of deep water (point A) is almost indistinguishable from MODIS reflectance products (since we do not have truth data, MODIS data were used as a reference in this article), which proves that the dark pixel method is effective. After the atmospheric correction, the energy of the visible bands was reduced by more than 90%.

In the L-S model, we select a series of SIPs to derive the water quality characteristics. In order to ensure the reliability of bathymetric inversion results, we first need to delimit the regions where reliable bathymetric inversion can be conducted and then randomly pick pixel points from these regions. We only performed the inversion on pixels where the spectral signal was distinguishable from that of deep water. Fig. 5(a) shows the spectral change along the transect FE, while Fig. 5(b) shows the details of roughly the first half of that transect. In Fig. 5(b), as the water depth decreases, the spectral signal begins to get stronger and the reflectivity of the blue band and the green band begins to increase. The red box in Fig. 5(b) highlights the spectrum of optically deep water; here, the spectral characteristics of C_i (18) do not respond to the change in water depth. Therefore, we set the values of the blue and green band reflectances at the right edge of the box as threshold 1 (black solid line), which is the maximum depth of the L-S model.

Typically, the error in simple atmospheric correction is about 10% [26], which in this case gives a reflectivity error of ±0.011 in the blue band and ±0.0067 in the green band. We set threshold 2 (black dotted line) at twice the atmospheric correction error to ensure that the signal from the bottom of water is strong enough. Because threshold 2 is larger

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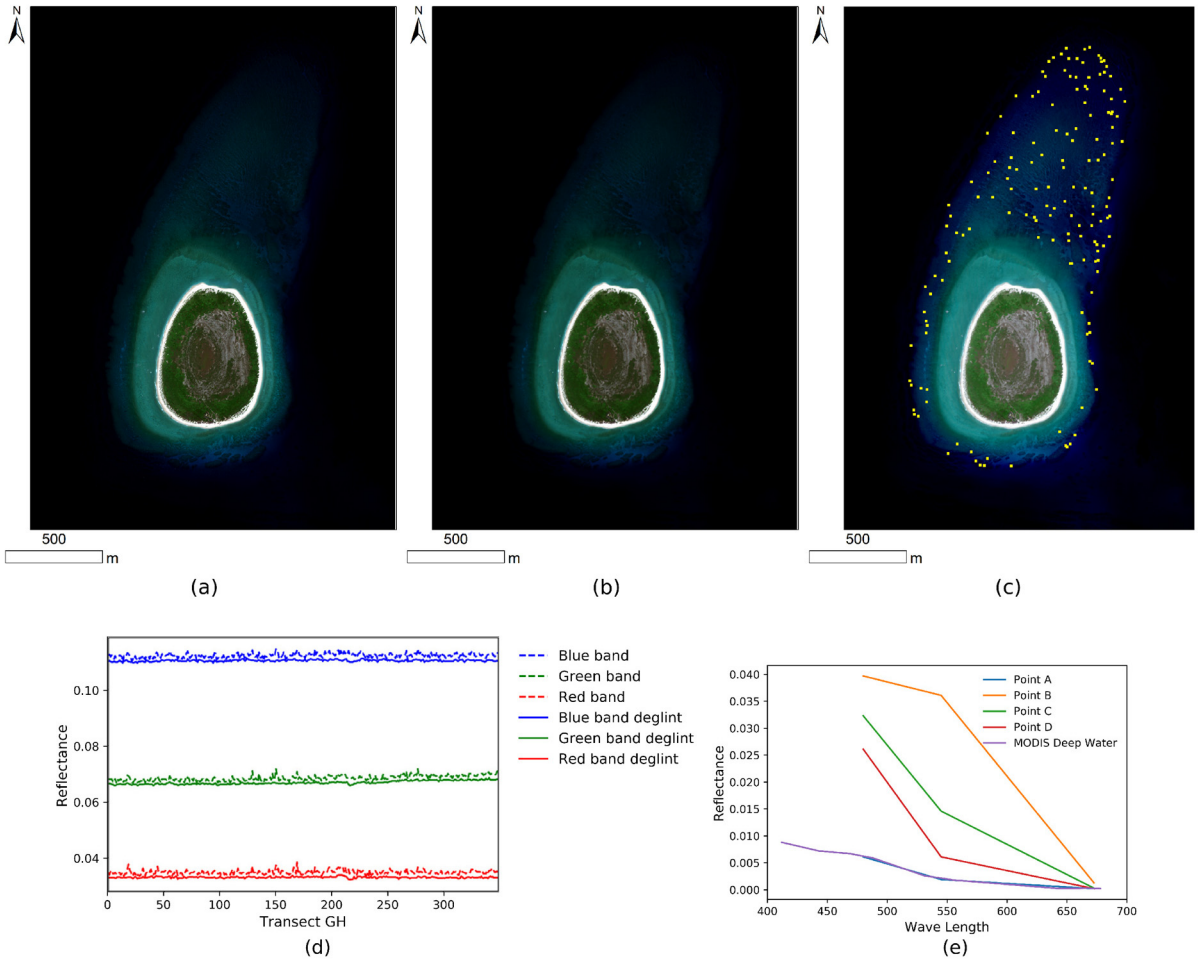


Fig. 4. (a) Image before sun glint correction. (b) Image after sun glint correction. (c) Image after sun glint correction and atmospheric correction; yellow points show the distribution of SIPs. (d) Changes in reflectivity before (dotted lines) and after (solids lines) sun glint correction. (e) Spectrum comparison after sun glint correction and atmospheric correction.

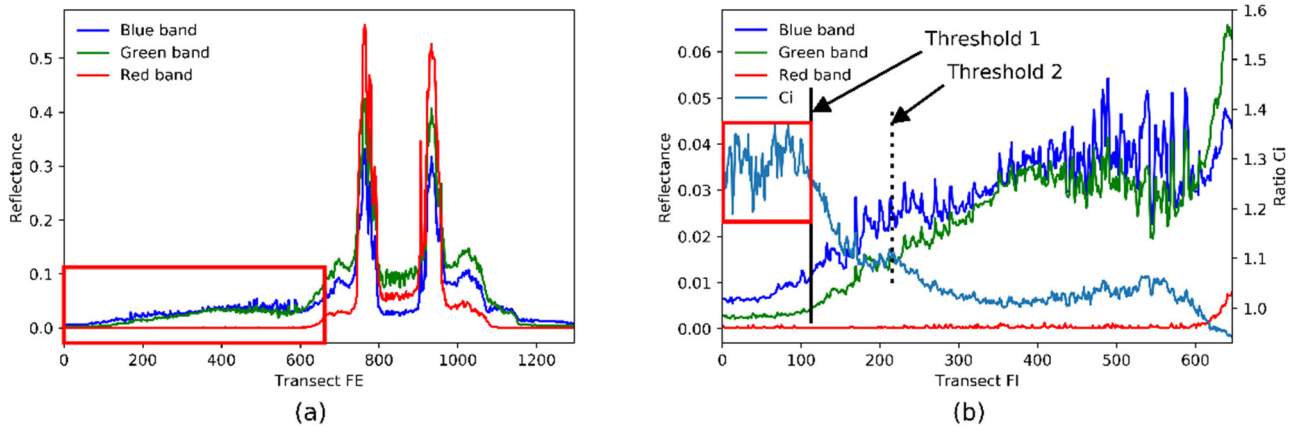


Fig. 5. (a) Reflectance along the transect FE in Fig. 3(b). (b) Details of the area outlined by the red box in (a).

than threshold 1, we select SIPs in the region determined by threshold 2.

Fig. 6(a) shows the results of the two inversion models. The scatter plot indicates that the results of the semianalytical model and the log-ratio model are mostly coincident within threshold 2 [dotted points in Fig. 6(a)], and that both models are accurate. However, when the test region is extended to the area bound by threshold 1 [indicated by the cross points in Fig. 6(a)], there is a significant difference between the two models. The results of the semianalytical model are lower than

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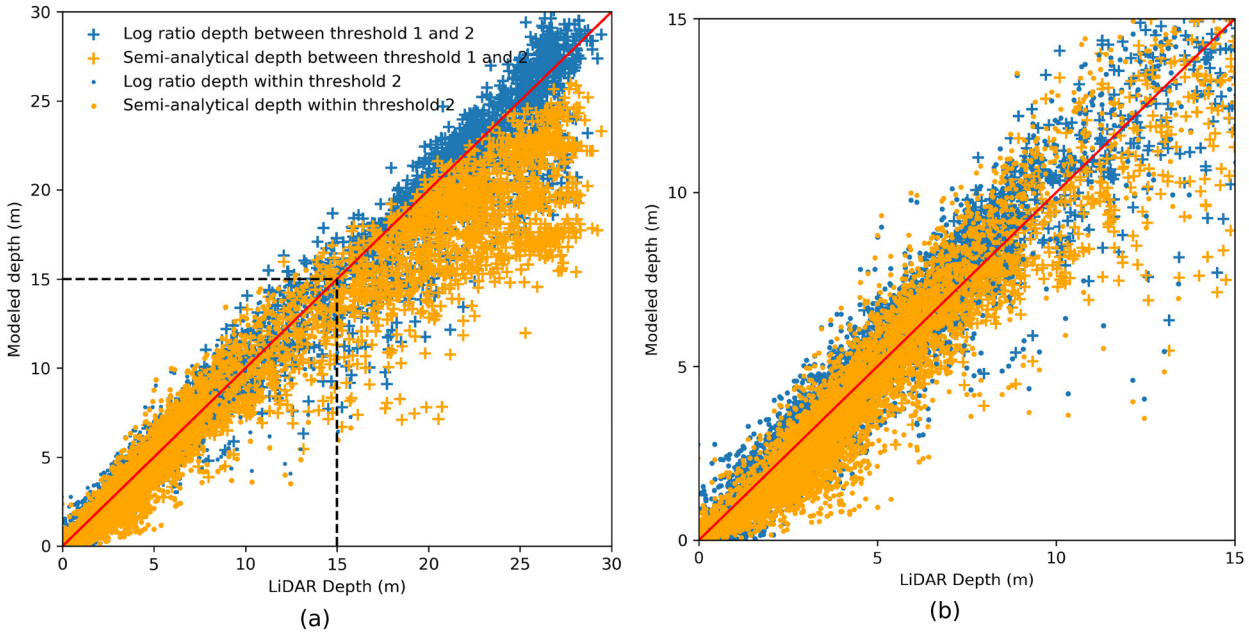


Fig. 6. (a) Results of two inversion models in the region with different thresholds in 0–30 m. (b) Results of two inversion models in the region with different thresholds in 0–15 m.

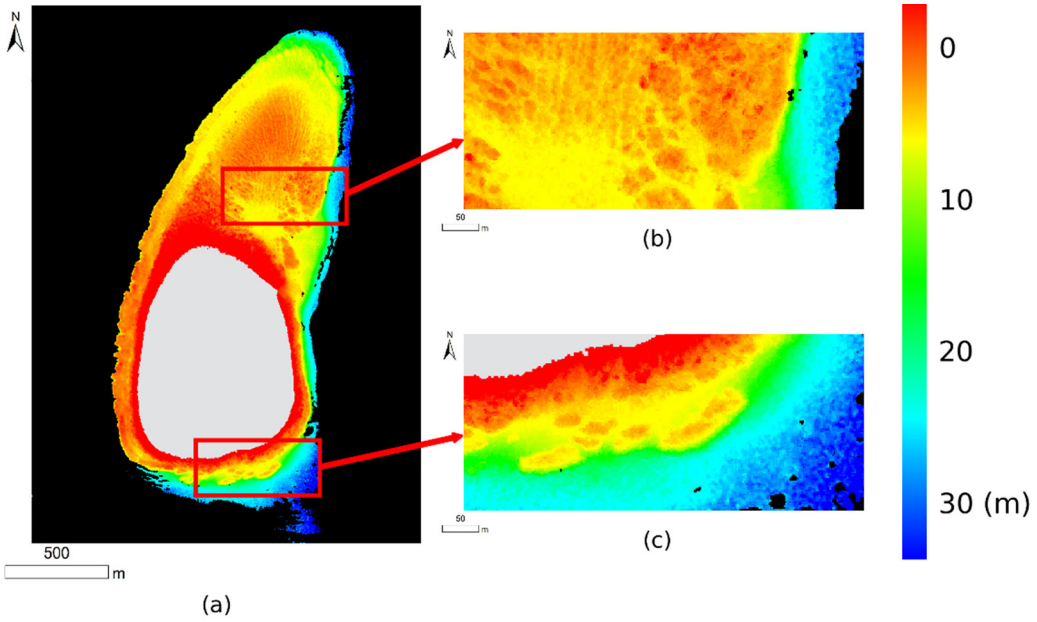


Fig. 7. (a) Calculated bathymetry of Ganquan Island. (b) and (c) Detailed insets of (a).

the reference LiDAR data, but those of the log-ratio model do not vary significantly from these truth data. In addition, the log ratio model is more computationally efficient. These two factors lead us to adopt the log-ratio model as the final output scheme of the L-S model. Fig. 6(b) shows a detailed scatter plot for 0–15 m.

The final results are shown in Fig. 7. The black areas represent deep-water areas where bathymetry cannot be retrieved, while the central gray region is the land area above the sea level. Fig. 7(b) and (c) shows the detailed structural features of underwater topography. The maximum depth that can be measured using this technique is approximately 30 m.

D. Tests Using WorldView-2, Sentinel-2, and Gaofen-1 Data

In order to further verify the quality of the L-S model using satellite data with different spatial resolutions and somewhat different wavelengths, we applied the same process on data from three other multispectral imagers—WorldView-2, Sentinel-2, and Gaofen-1. We used OSU Tidal Prediction Software (OTPS) (developed by Gary D. Egbert and Svetlana Y. Erofeeva from Oregon State University,

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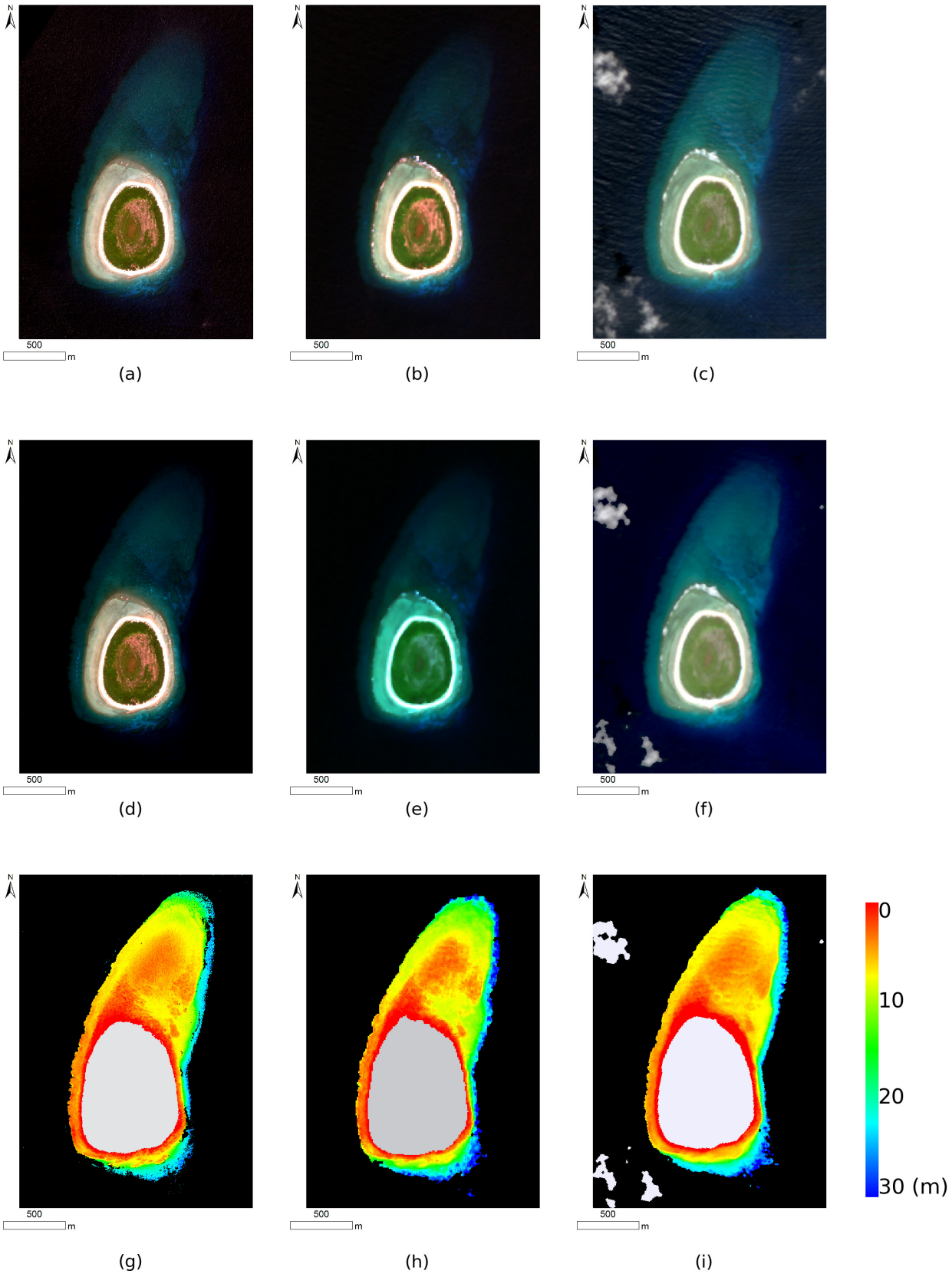


Fig. 8. True color images of Ganquan Island using (a) WorldView-2, (b) Sentinel-2, and (c) Gaofen-1 data. Color images after preprocessing using (d) WorldView-2, (e) Sentinel-2, and (f) Gaofen-1 data. Water depths calculated (g) GeoEye-1, (h) Sentinel-2, and (i) Gaofen-1 using the L-S method.

downloaded from http://volkov.oce.orst.edu/tides/) to calculate the tidal level corresponding to the imaging time of each scene. The information of the three satellites is shown in Table II.

Fig. 8 shows the results of the tests, in more detail; Fig. 8(a)–(c) shows raw remote sensing imagery, while Fig. 8(d)–(f) shows the imagery after preprocessing (including

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TABLE II
INFORMATION ON THE THREE SATELLITES

Satellite	Time	Resolution	Blue band	Green band	Red band	NIR band	Tide Level
WordlView-2	2014.11.08	2 m	480 nm	545 nm	660 nm	832.5 nm	-0.13 m
Sentinel-2	2018.06.19	10 m	490 nm	560 nm	665 nm	842 nm	0.42 m
Gaofen-1	2017.01.05	8 m	502 nm	576 nm	680 nm	810 nm	-0.17 m

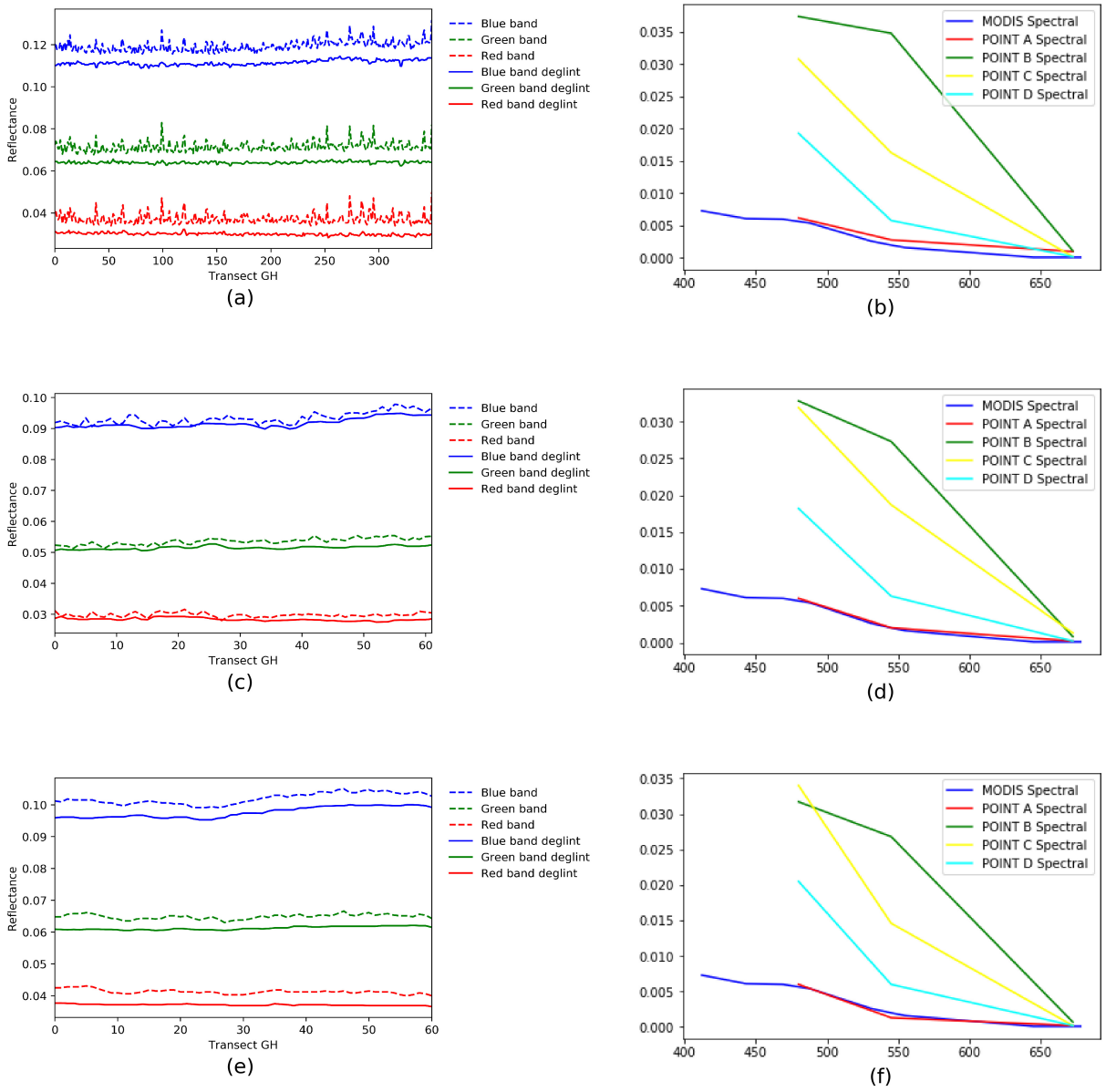


Fig. 9. (a), (c), and (e) Changes in reflectivity before (dotted lines) and after (solids lines) sun glint correction on WorldView-1, Sentinel-2, and Gaofen-1 sensors. (b), (d), and (f) Spectrum comparison after sun glint correction and atmospheric correction on WorldView-1, Sentinel-2, and Gaofen-1 sensors.

sun glint correction and atmospheric correction), and the final inversion results are shown in Fig. 8(g)–(i). Generally, the inversion results from different satellites have similar water depth gradients; however, due to the differences in sensor data quality and different environments during imaging, we found some differences in our article. The WorldView-2 results show more details, while the Sentinel-2 and Gaofen-1 results are coarser because of their different spatial resolutions and the quality of the inversion changed with depth. The results of this analysis are shown in Table III. The specific error will be discussed in Section IV.

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TABLE III
RMSE FOR DIFFERENT WATER DEPTHS

Model	Sensor	0-10 m		10-20 m		20-30 m		Overall			
		RMSE (m)	num	RMSE (m)	num	RMSE (m)	num	RMSE (m)	num	R2	Bias
L-S model	Geoeye-1	0.84	5706	2.28	746	2.36	1005	1.34	7458	0.97	0.10
	Worldview-2	0.90	5823	2.01	724	3.57	801	1.57	7353	0.95	-0.44
	Sentinel-2	0.98	5690	3.15	799	3.13	1080	1.82	7594	0.95	-0.28
	Gaofen-1	1.11	5519	3.54	697	3.51	1017	1.97	7234	0.93	-0.56
Log-ratio model	Geoeye-1	0.89	5706	2.29	746	2.28	1005	1.33	7458	0.97	0.01
	Worldview-2	0.95	5823	2.05	724	2.97	801	1.44	7353	0.96	-0.04
	Sentinel-2	0.99	5690	3.14	799	3.10	1080	1.78	7594	0.95	-0.28
	Gaofen-1	1.29	5519	3.13	697	3.08	1017	1.87	7234	0.94	-0.49

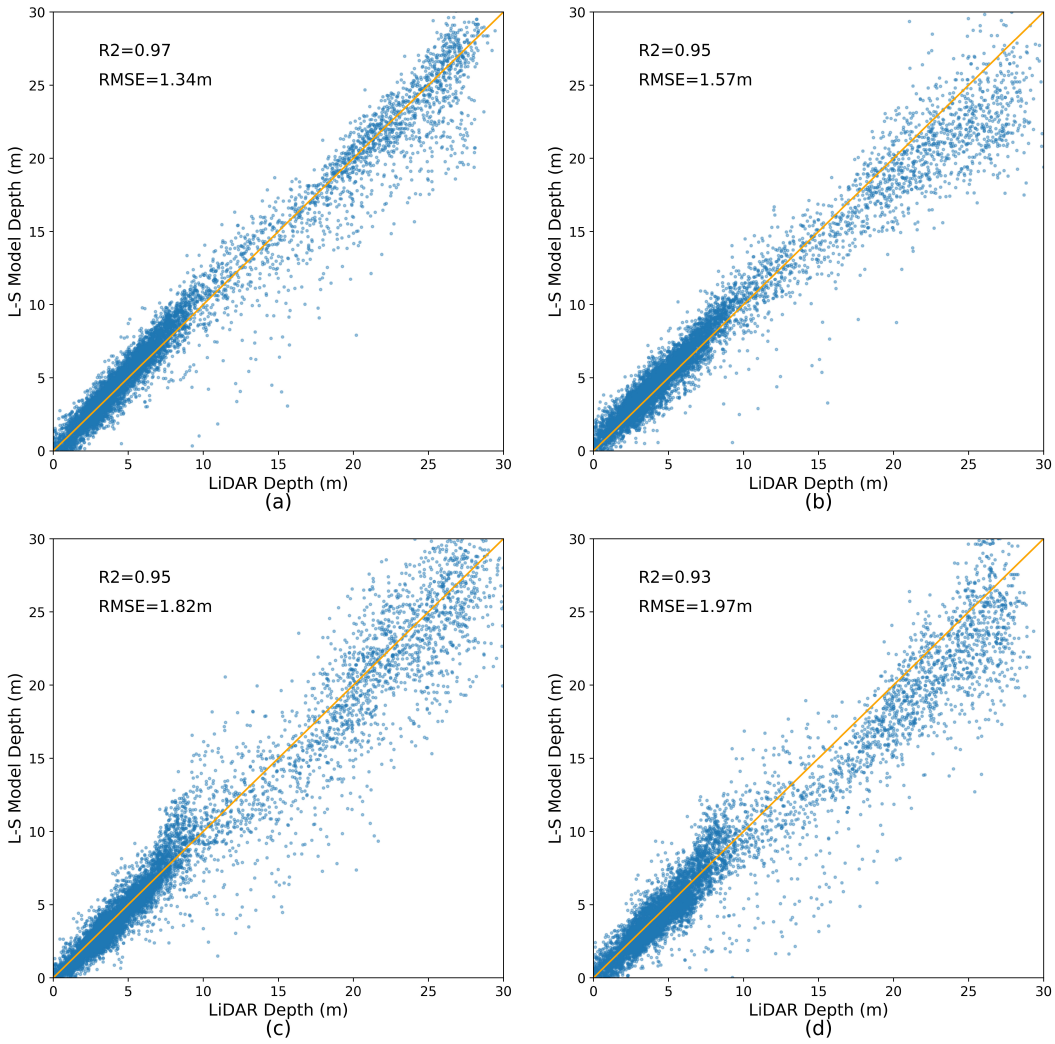


Fig. 10. Error scatter plot for (a) GeoEye-1, (b) WorldView-2, (c) Sentinel-2, and (d) Gaofen-1.

It is necessary to evaluate the performance of the atmosphere correction method for different sensors. Therefore, as shown in Fig. 9, we evaluated the spectral changes for different sensors before and after atmospheric and sun glint corrections. Furthermore, we used the optical deep-water spectrum of the MODIS products as the reference to verify

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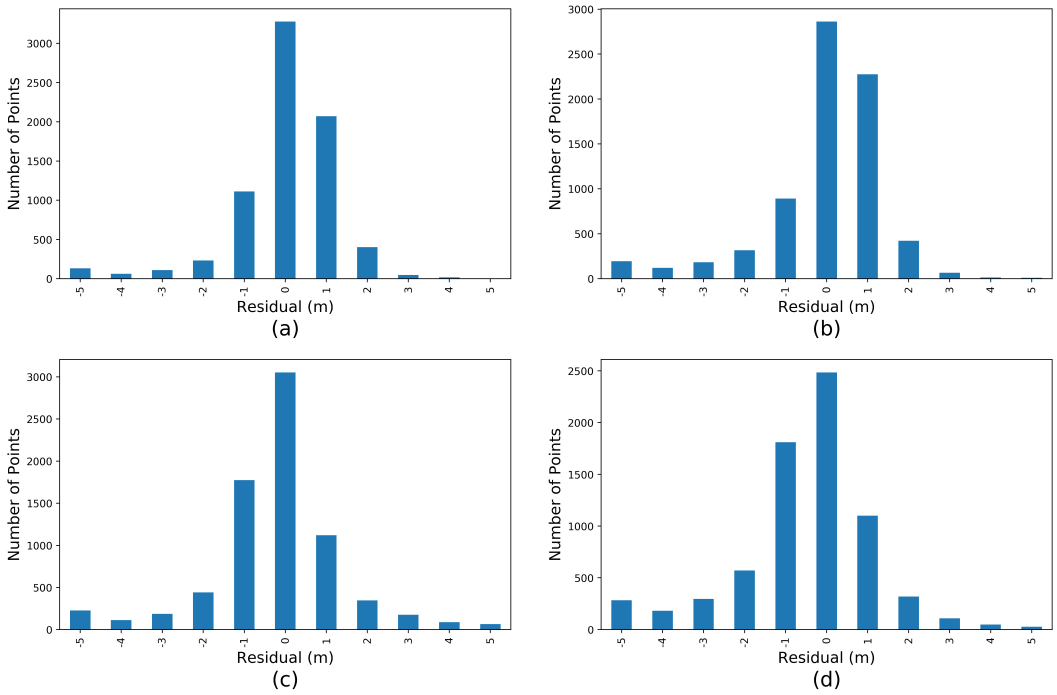


Fig. 11. Error distribution histograms of (a) GeoEye-1, (b) Worldview-2, (c) Sentinel-2, and (d) Gaofen-1 images.

the robustness of the atmospheric correction algorithm. Fig. 9(a), (c), and (e) shows the change before and after sun glint correction and Fig. 9(b), (d), and (f) shows the spectral change before and after atmospheric correction. In general, after atmospheric correction, the spectrum of the optically deep-water region of remote sensing imagery is adjusted to match the reflectivity of MODIS products for optically deep water; this proves the effectiveness of the dark pixel method. Further details on atmospheric correction are discussed in Section IV.

E. Numerical Results

To obtain more comprehensive results, we compared the results of the L-S model for different satellites and focused on the root-mean-square error (RMSE) at different water depths. Two hundred points were selected as control points to observe the corresponding indicators of the traditional log ratio model. As shown in Table III, we compared the performance of the two models (L-S model and log ratio model) for four satellite data, including the segmented RMSE, overall RMSE, overall R2, and bias.

Analysis of the results obtained from the four satellites are shown in Fig. 10. All water depth results were corrected for the tide level. In general, the L-S model produced accurate results for all four satellites. In terms of accuracy, the Geoeye-1 data have an RMSE value of 1.34 m and an R^2 value of 0.97; the Worldview-2 data have an RMSE value of 1.57 m and an R^2 value of 0.95; the Sentinel data have an RMSE value of 1.82 m and an R^2 value of 0.95, and the Gaofen-1 data have an RMSE value of 1.97 m and an R^2 value of 0.93. Thus, for the same satellite data, the L-S model produced results of accuracy similar to those by the algorithm presented in [19]. We also performed a segmented error analysis to determine whether four satellites are good in the range of 10 m, with an RMSE of approximately 1 m. As the water depth increases, the error also increases. In general, the errors for all satellites conform to the Gaussian distribution, as shown in Fig. 11, which indicates that this method is effective.

For comparison with the L-S model, we selected 200 LiDAR points from the region within threshold 1 for use in the traditional log-ratio model. The scatter plot of the results is shown in Fig. 12 and the segmented error is shown in Table II. The R^2 values for the GeoEye-1, WorldView-2, Sentinel-2, and Gaofen-1 data are all above 0.94 and RMSE values are less than 2 m. By comparing Fig. 10 with Fig. 12, we can see that the R^2 and RMSE values for the L-S model are very close to the log-ratio model and that we can achieve accurate inversion results without using *in situ* data to train the model.

IV. Discussion

A. Atmospheric Correction

Too much or too little atmospheric correction undoubtedly affects the accuracy of depth inversion. In this article, we adopted the dark pixel method (usually, the dark pixel comes from the shadow of clouds or ground occlusion), in which the energy of the atmosphere accounts for more than 90% of the energy received by the satellite. The log-ratio model has a strong anti-interference effect on the substrate and atmosphere, and the L-S model inherits this advantage, that is, compared with the semianalytical model, the L-S model has a stronger robustness for atmospheric correction.

B. Evaluation of Bathymetry Mapping Capability

In order to evaluate the bathymetry inversion capability of the L-S model, the L-S-modeled depth, LiDAR-measured

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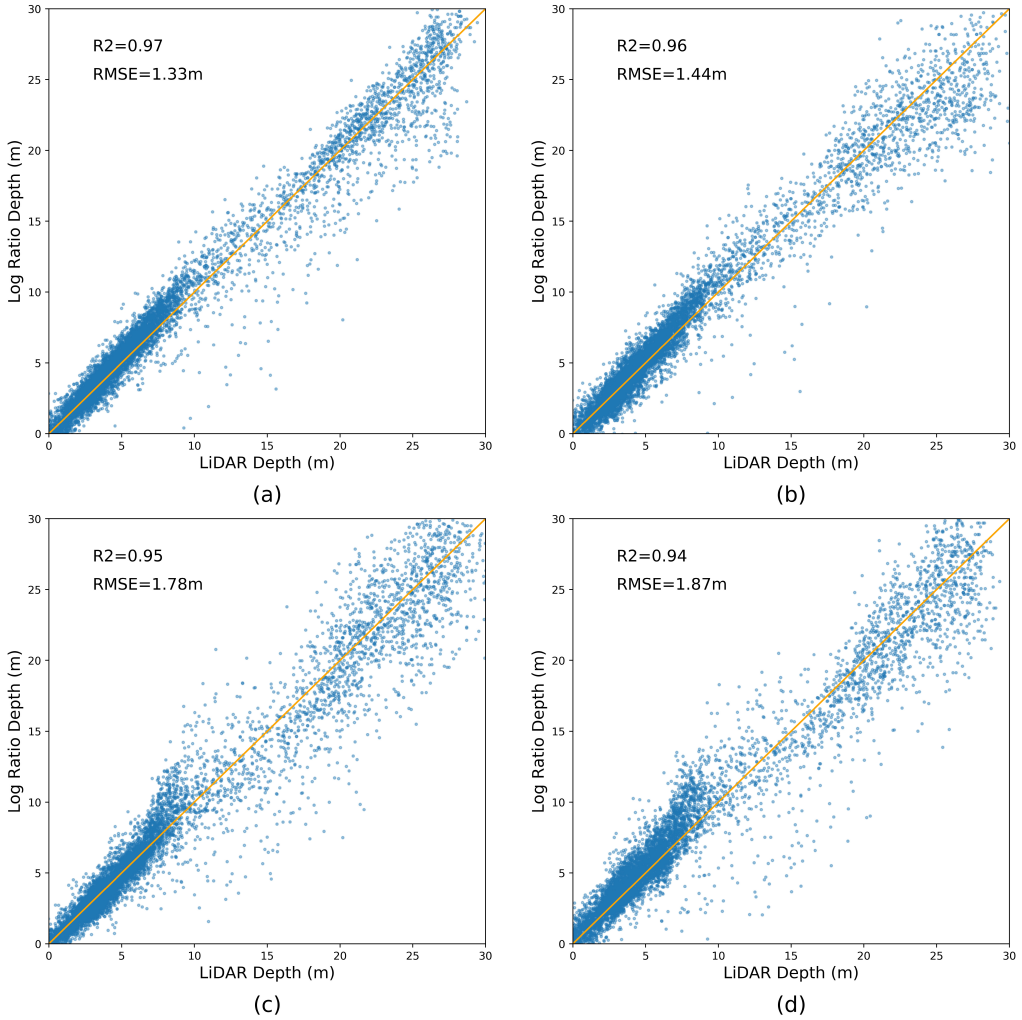


Fig. 12. Scatter plot of results using a traditional log-ratio model with (a) GeoEye-1, (b) Worldview-2, (c) Sentinel-2, and (d) Gaofen-1 data.

depth, and C_i were plotted for the transect FI, as shown in Fig. 5(b), the results of which are shown in Fig. 13. Fig. 13 shows that the LiDAR-measured depth gradually becomes shallower from a depth of 40 m, when the C_i is relatively unchanged, and the corresponding modeled depth is similar to C_i. Near threshold one—which occurs at a measured water depth of about 30 m—C_i begins to decline, and the inversion depth becomes shallower in line with the LiDAR-measured depth. Therefore, in order to obtain good bathymetric inversion results, it is reasonable and reliable to use the spectral characteristic C_i of water as the threshold rather than simply using bathymetric values to determine the inversion region.

We recommend setting threshold 2 at twice the atmospheric correction error to ensure that the signal from the bottom of water is strong enough, while the error in simple atmospheric correction is about 10% [26]. We recommend that the value of threshold 1 be set as the value where the reflectivity of blue and green bands tends to flatten, as shown in Fig. 13.

C. Effect of SIP Selection

In order to investigate the influence of SIP selection on the depth inversion results, we used GeoEye-1 data. Fig. 14 shows that as the number of pixel points increases, the final result improves. However, there are diminishing returns as the number of SIPs approaches 50, with the RMSE fluctuating within 0.05 m. It should be noted that the number is not fixed and will vary with environmental changes. With increasing environmental complexity, more sample points are necessary to ensure that these sample points reflect sufficient environmental information. Furthermore, in deeper water areas, the spectrum is more sensitive to atmospheric correction. If the atmospheric correction is insufficient, the inversion results in deeper regions are often low; if the atmospheric correction is excessive, the situation is opposite. However, the effect of atmospheric correction is not significant in the case of shallower waters (within threshold 2).

In this article, we also present a mathematical consequence to explain how may SIPs are enough. In this article, we made the following assumptions: 1) as per Stumpf [6], the model needs five control points and 2) since the semianalysis model needs to consider the substrate, we assumed that there are three substrates—sand, vegetation, and coral reef—and their area proportion is uniform. Therefore, the number of SIPs was multiplied by three from hypothesis I to 15. When the process is completely random, the expected number of SIPs can be

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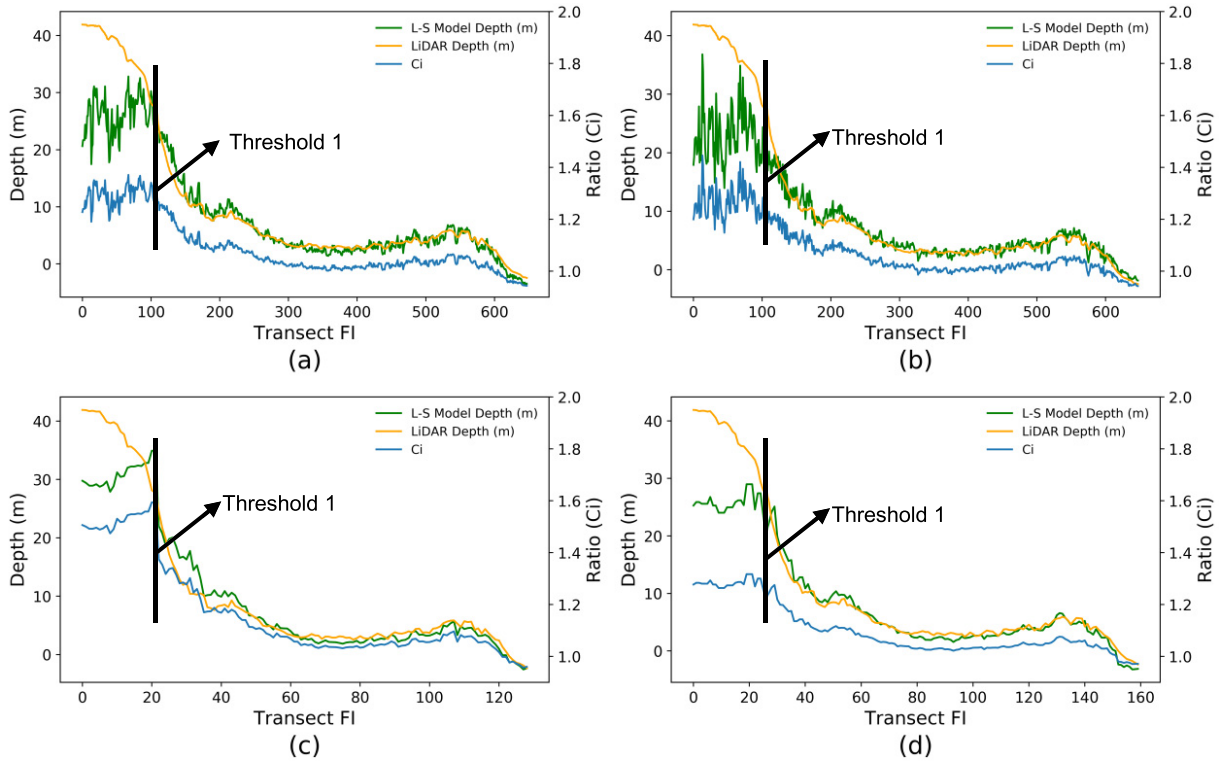


Fig. 13. LiDAR-measured depth, L-S model depth, and C_i of data from (a) GeoEye-2, (b) WorldView-2, (c) Sentinel-2, and (d) Gaofen-1.

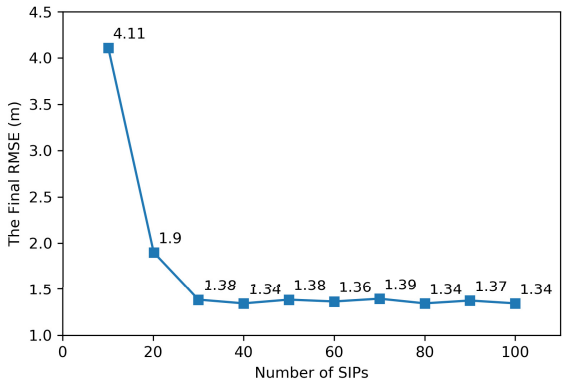


Fig. 14. Influence of the number of SIPs on the final result (RMSE).

calculated using the following formula:

$$N = f(x) = \sum_{i=1}^{x} \frac{x}{i} \tag{21}$$

where x is 15 and the number of random SIPs required is about 50.

D. Effect of Variation of Bottom Types

In the L-S model, the water depth and bottom albedo of SIPs can be simultaneously obtained under the assumption of homogeneity of water. Fig. 15 presents detailed information on the SIPs derived from GeoEye-1 images (as mentioned in Section III). In Fig. 15, different markers indicate different albedos and different colors indicate different water depths

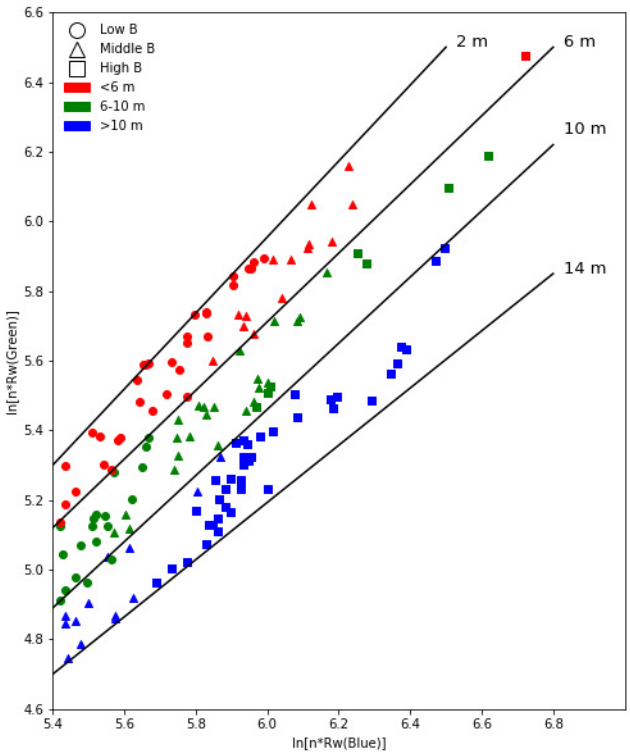


Fig. 15. Log transformation for the L-S model using the SIP data derived, as shown in Section III. The value of n is set to 10 000; the solid line shows the water depth derived using the log-ratio model; and different markers and colors represent different bottom albedos and water depths.

derived by the semianalytical model. The solid line shows the water depth derived using the log-ratio model. Fig. 15 is also consistent with the results presented in [6], which shows that

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there are different albedo types in each water depth range. Therefore, the L-S model is insensitive to the albedo-like log-ratio model [6].

V. Conclusion

This article proposes a new shallow water depth inversion method using four-band multispectral remote sensing images without the need for *in situ* training data. The most prominent feature of this model is that it establishes the connection between different pixels in a particular region, integrates the respective advantages of the semianalytical model and the log-ratio model, and logically achieves self-consistency and unity; it thus makes possible the realization of water depth inversion without the use of measured data when employing four-band imagery. The results show that high-resolution remote sensing images can achieve higher accuracy, because the influence of mixed pixels is low in such images, and the assumption of a constant water depth per pixel is more reliable.

In addition, our analysis of the impact of the SIP selection process on the results shows that the number of points being more than 50 is sufficient and reliable, and RMSE fluctuations of only about 0.05 m may be due to a random selection process. Compared to various types of sample selection processes required in [19], the random selection process for a small number of SIPs is required for the operation of the L-S model; the latter is more convenient. At the same time, SIP selection within the range of threshold 2 determined in this article can reduce the influence of atmospheric correction error and ensure higher accuracy of the water depth inversion results. Because only a limited number of SIPs are needed for the L-S model, and a simple calculation method can directly yield water depth inversion results corresponding to an entire image, the L-model has a very high computational efficiency. Our model inherits the advantages of the log-ratio model, which is insensitive to the bottom type [6], and it can be used to invert water depths with different albedo values B.

As an optical remote sensing method, the L-S model indicates that only those reflection signals from the seabed that are strong enough can yield effective bathymetry inversion results; thus, there are certain limitations in the case of turbid water bodies or low-reflectivity substrates.

In summary, the L-S method proposed here can achieve very close precision, compared to the traditional log-ratio model, in the absence of measured data. It can also use data from different types of sensors with different resolutions and is applicable to any type of area, because it is not an image-specific or a site-specific approach. Therefore, it is considered to be a remarkable progress in four-band multispectral remote sensing bathymetric inversion, providing a feasible method for water depth inversion without measured data.

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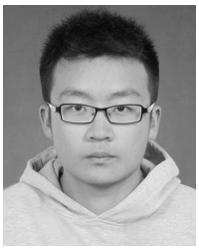
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